



Personal Habit Drift Detection and Forecasting System

A Major Project Report

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Abstract

People often fail to maintain consistent daily habits such as proper sleep, study or work routines, physical activity, and productivity. These habits may gradually improve or deteriorate over time without being noticed.

This project presents a machine learning-based system for detecting personal habit drift and forecasting future behavioral trends. The system analyzes daily behavioral data including sleep duration, sleep quality, study or work hours, exercise, screen time, mood, productivity, stress, caffeine intake, and water intake. A numerical Habit Score is calculated to represent the overall quality of daily habits. The Drift Score is then calculated from changes in the Habit Score to identify whether the user’s behavior is improving or declining.

The system uses a personal baseline to compare current behavioral values with the user’s normal historical pattern. Three regression models, namely Linear Regression, Random Forest Regression, and Gradient Boosting Regression, are evaluated for predicting the Drift Score. The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Based on the reported experimental results, Random Forest achieved the highest R^2 score of 0.1694, while Gradient Boosting achieved a very close R^2 score of 0.1643. Although Random Forest obtained the highest R^2 value, Gradient Boosting was selected as the final regression model for the proposed system because it provides strong predictive performance and is suitable for modeling non-linear relationships between behavioral factors and habit drift.

In addition to numerical prediction, the proposed system classifies habit drift into Improving, Stable, or Declining states. It also provides personal baseline comparison, future drift forecasting, early warning signals, and personalized habit insights. The system is intended for behavioral analysis and self-management and is not designed for medical diagnosis or clinical decision-making.

Keywords: Habit Drift, Machine Learning, Gradient Boosting, Personal Baseline, Behavioral Analysis, Forecasting, Early Warning

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Chapter 1

Introduction

1.1 Background

Human habits play an important role in daily life, academic performance, work, productivity, and general well-being. Habits such as sleep, study or work, physical activity, screen usage, mood, and productivity can change gradually over time. However, people may not notice these changes until they become significant.

Habit formation and behavioral consistency can vary considerably between individuals. Previous research has shown that habit development does not follow the same pattern for everyone, which supports the importance of considering individual behavioral patterns rather than using a single standard for all users [1].

Traditional habit-tracking systems mainly record daily activities and display historical statistics. Although this information is useful, it does not necessarily explain whether a person's habits are gradually improving or deteriorating. Historical charts also provide limited information about what may happen in the future.

The proposed Personal Habit Drift Detection and Forecasting System addresses this problem by analyzing daily behavioral patterns using machine learning. The system calculates a Habit Score from selected behavioral factors and compares behavioral observations with a personal baseline to identify changes from the user's normal pattern.

The system goes beyond simple tracking by detecting habit drift, predicting future drift, generating early warnings, and providing personalized explanations about changes in important behavioral factors.

1.2 Problem Statement

People's daily habits can gradually change without being noticed. For example, a person may slowly sleep less, study for fewer hours, exercise less, spend more time on screens, or experience changes in mood and productivity.

Traditional habit-tracking systems mainly record and display past activities. They generally do not analyze whether a person's current behavior is drifting from their own

normal behavioral pattern or provide predictive information about future changes.

As a result, users may recognize a negative behavioral pattern only after it has become significant.

Therefore, there is a need for a system that can analyze personal behavioral data over time, detect positive or negative habit drift, forecast possible future trends, identify repeated negative patterns, and provide understandable personalized insights.

1.3 Objectives

The main objective of this project is to develop a machine learning-based system that can detect personal habit drift and forecast future behavioral trends.

The specific objectives are:

- To collect and organize daily behavioral data related to personal habits.
- To calculate a numerical Habit Score from selected behavioral factors.
- To analyze changes in habits such as sleep, study or work, exercise, screen time, mood, stress, and productivity.
- To establish a personal baseline for each user based on historical behavioral data.
- To detect whether a user's habits are improving, stable, or declining.
- To use machine learning to predict the Drift Score.
- To forecast possible future habit drift using recent behavioral history.
- To provide early warnings when repeated negative behavioral patterns are detected.
- To provide simple personalized explanations about changes in the user's habits.

1.4 Scope of the Project

The system focuses on analyzing daily behavioral patterns and detecting changes in personal habits over time.

The main behavioral factors considered in this project include:

- Sleep duration
- Sleep quality
- Bedtime and wake-up time

- Study or work hours
- Productivity level
- Exercise duration
- Exercise type
- Screen time
- Screen activity
- Mood score
- Stress level
- Caffeine intake
- Water intake

The project includes:

- Daily habit data processing
- Habit Score calculation
- Drift Score calculation
- Machine learning-based prediction
- Personal baseline comparison
- Habit drift classification
- Future drift forecasting
- Early warning generation
- Personalized habit insights
- Visualization of habit trends

The system is intended for behavioral analysis and self-management. It is not designed for medical diagnosis, clinical decision-making, or treatment recommendations.

Chapter 2

Literature Review

Habit tracking and behavioral monitoring have become increasingly important with the development of digital technologies. Digital behavioral data can provide information about how an individual’s behavior changes over time and can support more personalized analysis and timely feedback [2].

Research on habit formation has shown that habits develop through repeated behavior and that the time required for habits to become automatic can vary considerably between individuals. Lally et al. studied habit formation in real-world settings and demonstrated considerable variation in habit development between individuals [1]. This variation supports the use of personalized behavioral baselines instead of applying exactly the same behavioral threshold to every user.

Traditional habit-tracking applications generally allow users to record activities such as exercise, sleep, mood, or productivity. These systems may provide historical charts and statistics, but recording data alone does not necessarily identify whether a user’s behavior is moving away from their usual pattern.

Machine learning provides an opportunity to analyze multiple behavioral variables simultaneously and identify relationships within behavioral data. Regression techniques are appropriate when the target is a continuous value such as a Drift Score. Ensemble methods such as Random Forest and Gradient Boosting can model non-linear relationships between input variables and target values [3].

Personalization is particularly important in behavioral analysis because normal behavior can differ between individuals. For example, a particular amount of sleep or study time may be normal for one person but represent a significant change for another person. Therefore, a personal baseline can provide a more individualized comparison.

Forecasting can further extend behavioral monitoring by estimating the possible future direction of a user’s behavioral pattern. Similarly, an early warning mechanism can identify repeated negative changes and provide feedback before the pattern becomes more significant.

Based on these concepts, the proposed system combines behavioral scoring, personal baseline analysis, machine learning prediction, habit drift classification, future forecasting, and early warnings into a single system.

Chapter 3

Methodology

The proposed system follows a sequence of steps to analyze personal behavioral patterns and detect habit drift. The overall methodology consists of data collection, data cleaning, preprocessing, Habit Score calculation, Drift Score calculation, feature selection, machine learning model training, model evaluation, personal baseline analysis, drift classification, forecasting, early warning generation, and personalized insights.

3.1 System Workflow

The overall workflow of the proposed system is:

Daily User Data → Data Cleaning → Habit Score → Drift Score → Feature Processing → ML Model → Prediction → Drift Detection → Forecasting → Warnings and Insights

3.2 Data Collection

The system uses a structured dataset containing daily behavioral information for multiple users. Each record represents the behavioral information of a user on a particular day.

The dataset contains variables related to sleep, study or work, exercise, screen usage, mood, productivity, stress, caffeine, and water intake.

The main dataset attributes are shown in Table 3.1.

Table 3.1: Main Behavioral Dataset Features

Feature	Description
User ID	Unique identifier of the user
Date	Date of the daily record
Sleep Duration	Number of hours slept
Sleep Quality	Self-reported sleep quality
Bedtime	Time at which the user went to sleep
Wake-up	Time at which the user woke up
Study/Work Hours	Daily study or work duration
Productivity Level	Self-assessed productivity level
Exercise Minutes	Duration of physical activity
Exercise Type	Type of exercise performed
Screen Time	Total daily screen usage
Screen Activity	Main type of screen activity
Mood Score	Self-reported mood level
Stress Level	Self-reported stress level
Caffeine Intake	Daily caffeine consumption
Water Intake	Daily water consumption
Habit Score	Calculated overall daily habit score
Drift Score	Change in Habit Score

3.3 Data Preprocessing

Before applying machine learning, the dataset is checked for missing values and inconsistent data.

Missing categorical values such as Exercise Type are handled by replacing missing values with an appropriate category such as “None”. Date values are converted into a consistent date format.

The records are also organized according to User ID and Date so that the behavioral history of each user can be analyzed in chronological order.

Categorical variables are converted into numerical representations using one-hot encoding where necessary. Variables that are not directly suitable for the selected machine learning models are removed or transformed into appropriate numerical features.

3.4 Habit Score Calculation

Machine learning models cannot directly understand concepts such as a “good day” or a “bad day”. Therefore, the system calculates a numerical Habit Score ranging from 0 to 100.

A higher Habit Score represents relatively better daily behavioral performance based on the selected behavioral indicators.

The current Habit Score is calculated using the following formula:

$$\begin{aligned} HabitScore = & \left(\frac{SleepDuration}{10} \times 20 \right) + \left(\frac{StudyWork}{12} \times 15 \right) \\ & + \left(\frac{Productivity}{10} \times 15 \right) + \left(\frac{Exercise}{240} \times 15 \right) \\ & + \left(\frac{Mood}{10} \times 15 \right) + \left(\frac{10 - Stress}{10} \times 10 \right) \\ & + \left(\frac{13 - ScreenTime}{12} \times 10 \right) \end{aligned} \quad (3.1)$$

The maximum contribution of the seven components is:

$$20 + 15 + 15 + 15 + 15 + 10 + 10 = 100 \quad (3.2)$$

Therefore, the maximum theoretical Habit Score is 100.

The formula uses positive contributions for desirable behavioral factors and reverse scoring for factors where higher values are considered undesirable, such as stress and screen time.

The ranges used in the formula represent predefined behavioral reference ranges established for the project. They are used as scoring assumptions and should not be interpreted as clinical health standards.

3.5 Drift Score Calculation

The Drift Score represents the change in daily habit quality compared with the previous observation.

The project uses the following formula:

$$DriftScore_t = HabitScore_t - HabitScore_{t-1} \quad (3.3)$$

For example, if the previous day’s Habit Score is 80 and the current day’s score is 60:

$$DriftScore = 60 - 80 = -20 \quad (3.4)$$

The negative value indicates that the overall Habit Score decreased. Similarly, if the Habit Score increases from 60 to 75:

$$DriftScore = 75 - 60 = 15 \tag{3.5}$$

This indicates an improvement.

It is important to distinguish this Drift Score from a direct measure of deviation from an ideal healthy lifestyle. The project's Drift Score measures the *change in behavioral score over time*.

3.6 Feature Selection

The machine learning models use selected behavioral variables as input features. The main features include:

- Sleep Duration
- Sleep Quality
- Study or Work Hours
- Productivity Level
- Exercise Duration
- Screen Time
- Mood Score
- Stress Level
- Caffeine Intake
- Water Intake

The target variable is the Drift Score. Therefore:

$$X = \text{Behavioral Features} \tag{3.6}$$

$$y = DriftScore \tag{3.7}$$

User ID and Date are not directly used as numerical predictive features.

3.7 Machine Learning Models

The project evaluates three regression models:

1. Linear Regression
2. Random Forest Regression
3. Gradient Boosting Regression

The purpose of comparing multiple models is to determine which model provides strong predictive performance for the available behavioral dataset and is suitable for modeling non-linear behavioral relationships.

3.7.1 Linear Regression

Linear Regression is used as a simple baseline model. It attempts to learn a linear relationship between the behavioral features and the Drift Score.

Its main advantages are simplicity, fast training, and easy interpretation. However, it may not capture complex non-linear relationships between behavioral factors.

3.7.2 Random Forest Regression

Random Forest Regression combines predictions from multiple decision trees. Each tree learns different relationships from the training data, and the final prediction is produced by aggregating the individual tree predictions.

Random Forest is useful for structured tabular data because it can capture non-linear relationships and interactions between features.

3.7.3 Gradient Boosting Regression

Gradient Boosting builds an ensemble of decision trees sequentially. Each new tree attempts to reduce the errors made by the previous trees.

This allows Gradient Boosting to model complex relationships between behavioral variables and Drift Score. Because of its strong predictive performance and suitability for non-linear behavioral relationships, Gradient Boosting is selected as the final regression model for the proposed system.

3.8 Train-Test Split

The dataset is divided into training and testing subsets using an 80:20 ratio.

The training data is used to learn the relationship between behavioral features and Drift Score, while the testing data is used to evaluate the trained models on previously unseen observations.

The model training process can be represented as:

$$Model.fit(X_{train}, y_{train}) \quad (3.8)$$

After training, predictions are generated using:

$$\hat{y} = Model.predict(X_{test}) \quad (3.9)$$

3.9 Model Evaluation

The models are evaluated using four metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

MAE measures the average absolute prediction error:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3.10)$$

MSE gives greater importance to larger prediction errors:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3.11)$$

RMSE is the square root of MSE:

$$RMSE = \sqrt{MSE} \quad (3.12)$$

The coefficient of determination is:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (3.13)$$

For MAE, MSE, and RMSE, lower values indicate better performance. For R^2 , a higher value indicates better explanatory performance.

3.10 Personal Baseline

To make the system personalized, a baseline is calculated for each user.

The personal baseline represents the user's typical behavioral level based on historical observations.

For a selected behavioral feature f , the baseline can be represented as:

$$Baseline_f = \frac{1}{n} \sum_{i=1}^n f_i \quad (3.14)$$

The latest value is compared with the corresponding baseline:

$$Deviation_f = Current_f - Baseline_f \quad (3.15)$$

This allows the system to identify whether a user's current behavior is higher or lower than their normal historical level.

3.11 Habit Drift Detection

The system converts the Drift Score into a simple behavioral status.

The three possible states are:

- **Improving**
- **Stable**
- **Declining**

A positive Drift Score indicates improvement, a negative Drift Score indicates decline, and values close to zero indicate stability.

The exact classification thresholds are predefined by the system.

3.12 Future Forecasting

The forecasting component uses recent Drift Score observations to estimate the next Drift Score.

The previous three Drift Scores are used as input values:

$$Forecast_t = f(Drift_{t-1}, Drift_{t-2}, Drift_{t-3}) \quad (3.16)$$

The predicted value is then converted into a meaningful behavioral status such as Improving, Stable, or Declining.

The forecasting component is intended to provide an indication of possible future direction rather than a medical or clinical prediction.

3.13 Early Warning System

The early warning component examines recent Drift Score observations to identify repeated negative patterns.

If multiple recent observations show significant negative drift, the system generates an early warning.

For example, if the system observes repeated decreases in Drift Score, it may display a message indicating that the user's behavioral pattern has been declining recently.

The warning system therefore adds an interpretable layer beyond the numerical machine learning prediction.

3.14 Personalized Insights

Finally, the system compares the user's latest behavioral values with their personal baseline.

For example, if screen time is substantially higher than the user's normal level while study time and sleep duration are lower, the system can identify these changes and provide a simple explanation.

The system may generate insights such as:

- Study time has decreased compared with the personal baseline.
- Screen time is higher than the usual level.
- Recent Drift Scores indicate a declining pattern.
- Recent behavior suggests improvement in physical activity.

These insights are intended to help users understand their behavioral patterns rather than simply presenting machine learning numbers.

Chapter 4

Experimental Setup and Results

4.1 Experimental Setup

The dataset is divided into training and testing sets using an 80:20 ratio. The training data is used to train the regression models, while the testing data is used to evaluate their predictive performance on unseen observations.

Three regression models are evaluated in this project: Linear Regression, Random Forest Regression, and Gradient Boosting Regression.

The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

4.2 Model Performance

Table 4.1 presents the performance of the three evaluated regression models.

Table 4.1: Performance of Regression Models

Model	MAE	MSE	RMSE	R^2
Linear Regression	5.90	57.94	7.61	0.1283
Random Forest	6.21	59.10	7.69	0.1694
Gradient Boosting	6.52	68.51	8.28	0.1643

The results show that Random Forest achieved the highest R^2 score of 0.1694. Gradient Boosting achieved a very close R^2 score of 0.1643, while Linear Regression achieved 0.1283.

Although Random Forest achieved the highest numerical R^2 value, the performance difference between Random Forest and Gradient Boosting is relatively small. Gradient Boosting was selected as the final model for the proposed system because it provides strong predictive performance and is suitable for modeling non-linear relationships between behavioral factors and habit drift.

The comparison also demonstrates that ensemble-based models can capture behavioral relationships that may not be fully represented by a simple linear model.

Therefore, Gradient Boosting Regression is used as the final regression model for the prediction and analysis components of the proposed system.

4.3 Prediction Performance

Figure 4.1 presents the relationship between the actual Drift Scores and the predicted Drift Scores generated by the selected Gradient Boosting model.

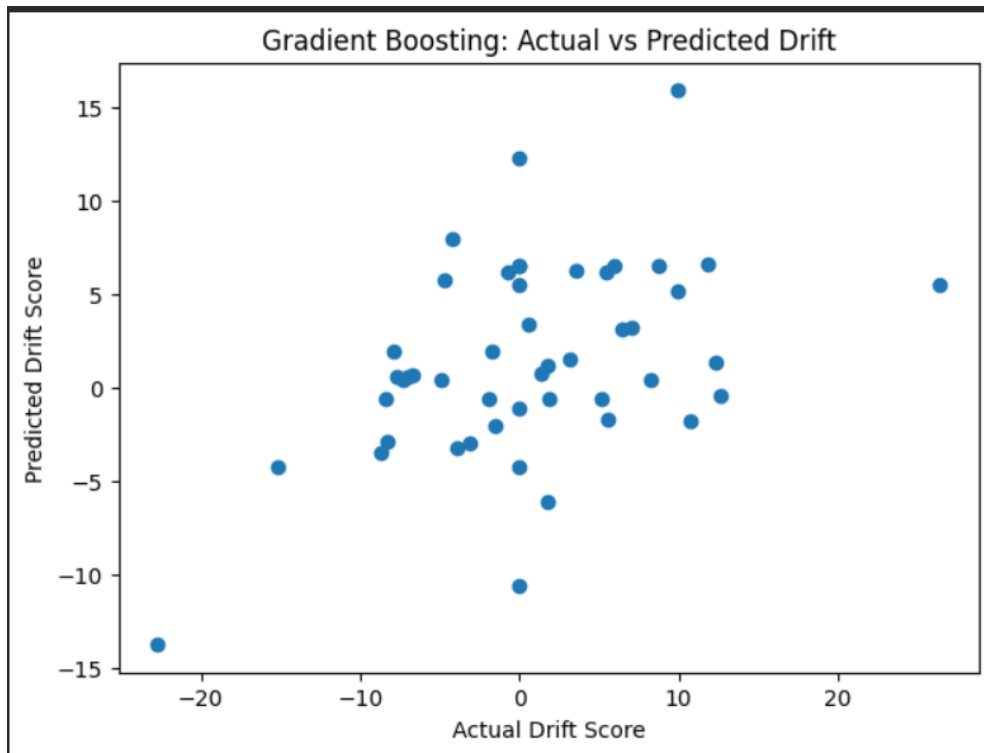


Figure 4.1: Actual vs Predicted Habit Drift Score

The figure provides a visual comparison between the actual Drift Scores and the values predicted by the Gradient Boosting model. Predictions that remain closer to the corresponding actual values indicate better prediction performance.

The figure helps evaluate how closely the model's predictions follow the observed Drift Score values.

4.4 Feature Importance Analysis

Feature importance is used to understand which behavioral variables contribute more strongly to the predictions made by the Gradient Boosting model.

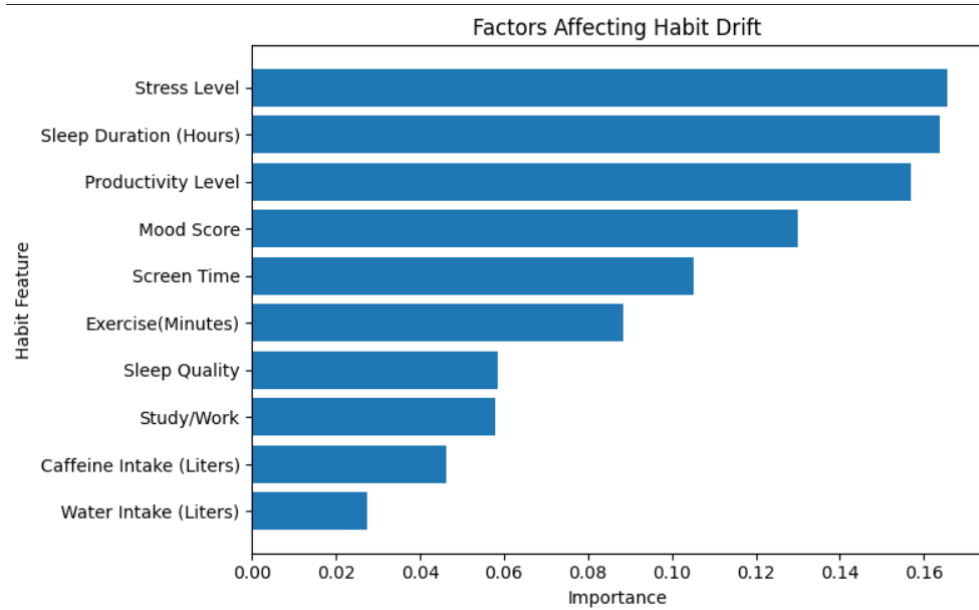


Figure 4.2: Importance of Behavioral Features in Predicting Habit Drift

The feature importance results provide an interpretation of the machine learning model by showing which behavioral factors had a greater influence on the prediction.

For example, if stress, sleep duration, or screen time receives a higher importance value, it indicates that the Gradient Boosting model relied more strongly on those variables when predicting the Drift Score.

This analysis helps transform the numerical prediction into more understandable behavioral information for the user.

4.5 Personal Habit Drift Analysis

The system analyzes the user's Drift Score over time. This allows the user to observe whether the overall behavioral pattern is generally improving, stable, or declining.

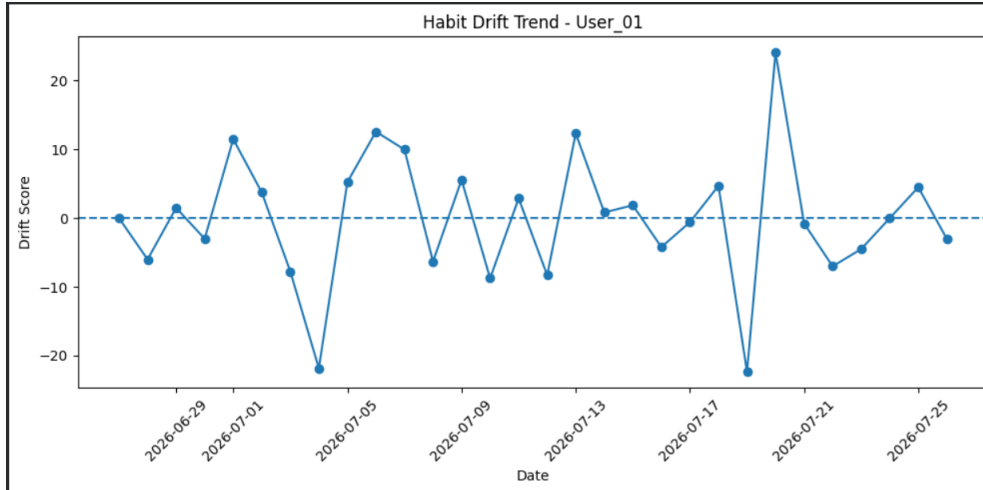


Figure 4.3: Personal Habit Drift Trend Over Time

The trend provides a visual representation of changes in the user's behavior across different observations. Positive Drift Score values represent improvement in the Habit Score, while negative values represent a decrease.

Repeated negative Drift Scores may indicate that the user's recent behavioral pattern is moving in a declining direction.

4.6 Future Habit Drift Forecast

The forecasting component uses recent Drift Score observations to estimate the next Drift Score and provide an indication of the possible future behavioral direction.

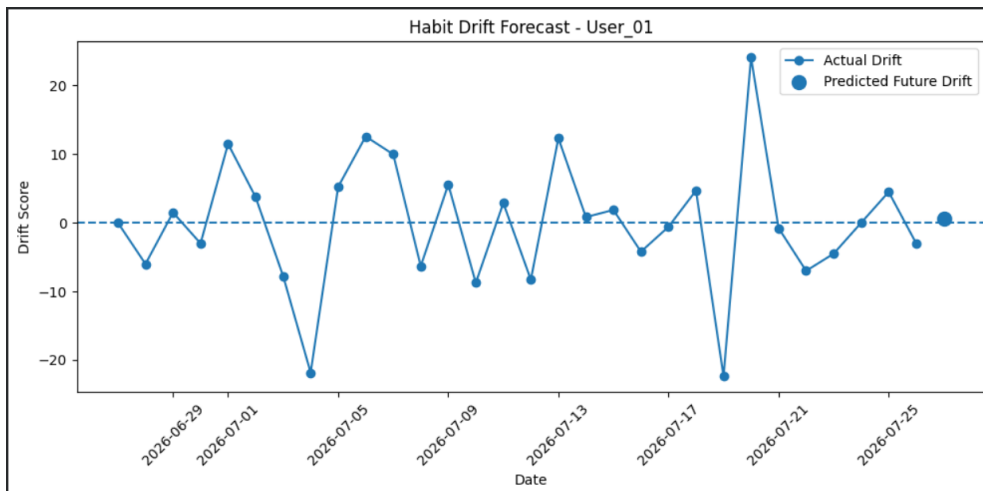


Figure 4.4: Future Habit Drift Forecast

The forecast provides an estimated future Drift Score based on the available recent behavioral pattern. The predicted value can also be converted into a simple behavioral status such as Improving, Stable, or Declining.

The forecasting result is intended to provide an early indication of possible behavioral changes rather than an exact long-term prediction.

4.7 Early Warning Results

The early warning component examines recent Drift Score observations to identify repeated negative behavioral patterns.

When multiple recent observations indicate significant negative changes, the system generates an early warning. This provides an additional layer of personalized feedback beyond the numerical machine learning prediction.

For example, if the user's Drift Score remains negative across several recent observations, the system may generate a warning indicating that the user's recent behavioral pattern is moving in a negative direction.

The purpose of the warning system is to help users notice potentially declining patterns earlier and encourage them to review the behavioral factors contributing to the change.

Chapter 5

Discussion

The experimental results demonstrate that machine learning can be used to analyze behavioral factors and predict changes in a user’s habit-related score.

Three regression models were evaluated: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. Random Forest achieved the highest R^2 score of 0.1694, while Gradient Boosting achieved a very close score of 0.1643. Linear Regression achieved an R^2 score of 0.1283.

Although Random Forest achieved the highest R^2 value, the difference between Random Forest and Gradient Boosting was relatively small. Gradient Boosting was selected as the final model for the proposed system because it provides strong predictive performance and is well suited to modeling non-linear relationships between multiple behavioral factors and the Drift Score.

The relatively low R^2 values indicate that the current dataset contains substantial variability that is not fully explained by the selected behavioral features. Therefore, the results should be interpreted as an initial demonstration of the proposed approach rather than as a highly accurate behavioral prediction system.

An important part of the project is that the system does not only produce a machine learning prediction. It also compares current behavior with a personal baseline. This allows the system to identify changes relative to the individual’s own historical pattern.

For example, a user may normally study for six hours per day. If the user’s recent study time decreases to three or four hours, the system can identify the change relative to that user’s normal behavior.

The early warning component uses recent Drift Scores to identify repeated negative changes. The forecasting component further attempts to estimate the next Drift Score from recent observations.

Together, these components allow the system to provide more understandable feedback instead of presenting only a numerical machine learning prediction.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

The Personal Habit Drift Detection and Forecasting System is a machine learning-based project designed to analyze changes in a person’s daily behavioral patterns and identify whether their habits are improving, remaining stable, or declining. The main purpose of the system is to go beyond simple habit tracking by comparing a user’s current behavior with their personal baseline and identifying gradual changes in daily habits.

The system processes daily behavioral factors such as sleep duration, sleep quality, study or work hours, exercise, screen time, mood, productivity, stress, caffeine intake, and water intake. These behavioral factors are converted into numerical information through Habit Score calculation. The Drift Score is then calculated from changes in the Habit Score to identify improvements or declines in daily behavioral patterns.

Three regression models, namely Linear Regression, Random Forest Regression, and Gradient Boosting Regression, were evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

The experimental results showed that Random Forest achieved the highest R^2 score of 0.1694, while Gradient Boosting achieved a very close score of 0.1643. Although Random Forest achieved the highest numerical R^2 value, Gradient Boosting was selected as the final model because it provides strong predictive performance and is suitable for modeling non-linear relationships between behavioral factors and habit drift.

In addition to machine learning prediction, the proposed system includes personal baseline analysis, habit drift classification, future drift forecasting, early warning detection, and personalized insights. These components help transform numerical machine learning results into information that can be more easily understood by users.

For example, repeated negative Drift Scores can indicate that a user’s recent behavioral pattern is declining. The system can then generate an early warning and provide information about behavioral factors that may be contributing to the change.

Overall, the project demonstrates how machine learning can be combined with personalized behavioral analysis to detect gradual habit changes and provide an indication

of their possible future direction. The system therefore provides a more personalized approach than a traditional habit-tracking system while focusing on behavioral analysis and self-management rather than medical or clinical diagnosis.

6.2 Limitations

The proposed system has several limitations.

First, the dataset used in the current study is relatively small, which may limit the generalizability of the machine learning models to a larger and more diverse population.

Second, some behavioral variables, such as mood and productivity, are based on user-provided ratings. Therefore, these values may be subjective and can vary depending on the user’s perception.

Third, the current Habit Score depends on predefined scoring ranges. These ranges are project-specific assumptions and do not represent universal medical or psychological standards.

Fourth, the Drift Score represents the change in Habit Score from one observation to another. Therefore, it measures short-term behavioral change rather than providing a complete measure of long-term behavioral quality.

Fifth, the forecasting component uses a limited number of previous Drift Score observations to estimate future drift. Longer historical sequences and dedicated time-series models may provide more robust forecasting in future versions.

Finally, the system is intended for behavioral analysis and self-management. It should not be considered a medical diagnostic or clinical decision-making system.

6.3 Future Work

The proposed system can be improved in several ways in future work:

- Collecting larger datasets from more users to improve the generalizability of the machine learning models.
- Collecting behavioral data over longer periods to capture long-term habit patterns.
- Using chronological time-series train-test splitting to better represent real-world future prediction scenarios.
- Evaluating advanced time-series models such as LSTM and other sequential learning approaches.
- Improving the personalization of baseline calculations based on individual behavioral patterns.

- Using rolling and moving statistical features such as 7-day averages, 14-day averages, trend slopes, and habit variance.
- Adding direction accuracy as an additional evaluation metric to measure whether the system correctly predicts improvement or decline.
- Improving the explanation layer using advanced explainability techniques such as SHAP.
- Developing a real-time habit monitoring application that can automatically collect and analyze daily behavioral information.
- Integrating wearable or mobile-device data where appropriate to reduce dependence on manual data entry.
- Providing more detailed and adaptive recommendations based on the user's individual behavioral history.
- Evaluating the complete system using a larger and more diverse behavioral dataset.

Chapter A

Appendix

This appendix can contain additional materials that are not necessary in the main body of the report but may be useful for understanding the implementation.

A.1 Example Machine Learning Pipeline

The main machine learning pipeline can be summarized as:

1. Load the dataset.
2. Check and handle missing values.
3. Convert date values into a consistent format.
4. Process categorical variables.
5. Calculate the Habit Score.
6. Calculate the Drift Score.
7. Select input features and target variable.
8. Divide the data into training and testing sets.
9. Train the regression models.
10. Generate predictions.
11. Calculate evaluation metrics.
12. Compare model performance.
13. Select the final model.
14. Generate behavioral insights.

A.2 Example Feature and Target Definition

The machine learning input can be represented as:

$$X = [SleepDuration, SleepQuality, StudyWork, Productivity, Exercise, ScreenTime, Mood, Stress, \dots] \quad (A.1)$$

The target is:

$$y = DriftScore \quad (A.2)$$

A.3 Example Interpretation

Suppose a user's recent behavioral records show:

- Decreasing study time
- Increasing screen time
- Decreasing exercise
- Lower mood
- Higher stress

If these changes result in decreasing Habit Scores, the Drift Score may become negative. Repeated negative values can cause the system to classify the user's recent pattern as Declining and generate an early warning.

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